

Code	Title	Sem.	Credits	Lec	PS	Lab
	Applied Mathematics & Statistics for Computational Sciences	M1S1	5	14	20	16

Description

This course provides a comprehensive foundation in the essential mathematical and statistical concepts that constitute the modern computational sciences, with a particular focus on their application in Artificial Intelligence, Machine Learning and other disciplines. Students will delve into key topics such as linear algebra, multivariate calculus, optimization techniques, and probability theory. Through a blend of theoretical understanding and hands-on practical sessions using Python libraries like NumPy, SciPy, Pandas, and Matplotlib, the course aims to equip students with the analytical tools and computational skills necessary to effectively implement, understand, and interpret advanced AI and ML methods.

Content

1. Introduction and Mathematical Foundations

- Overview of applied mathematics in AI and ML
- Refresher: key algebraic foundations, functions, and equations
- Vector spaces (definitions and properties)
- Practical session: Review Python programming and essential mathematical libraries (NumPy, SciPy)

2. Linear Algebra for Machine Learning

- Vectors and matrices
- Matrix arithmetic
- Linear transformations
- Eigenvalues and eigenvectors
- Matrix decompositions (e.g., Eigen-decomposition, Singular Value Decomposition - SVD)
- Application: Principal Component Analysis (PCA)
- Practical sessions: Python exercises on matrix manipulation, linear transformations, and PCA (NumPy, Scikit-learn)

3. Calculus and Optimization Essentials for ML

- Introduction to Limits, Continuity, and Derivatives
- Partial derivatives and gradients
- Multivariate calculus and optimization basics
- Optimization techniques: Gradient Descent (batch, stochastic, mini-batch), Newton's method
- Convex optimization
- Practical sessions: Implementing gradients and optimization algorithms in Python (hands-on Lab with examples)

4. Probability Theory and Distributions

- Probability spaces, random variables and events
- Conditional probability, Bayes' theorem, and independence
- Probability distributions (Discrete vs. Continuous)

- Distributions relevant for Machine Learning: Binomial, Normal, Gaussian, Bernoulli, Uniform, Poisson
- Central Limit Theorem
- Practical sessions: Probability simulation, sampling exercises, visualization and exploration of distributions with Python (NumPy, Matplotlib, SciPy)

5. Statistical Inference and Analysis

- Sampling distributions, confidence intervals, and hypothesis tests
- Parametric vs. Non-parametric tests
- Simple and Multiple Linear Regression
- Correlation and covariance analysis
- Metrics for model evaluation (R^2 , RMSE, MAE)
- Practical sessions: Statistical tests, regression, and correlation exercises with Python (Scipy.stats, Statsmodels, Scikit-learn)

6. Introduction to Bayesian Methods and Course Wrap-up

- Bayesian Probability and statistics
- Prior, likelihood, posterior distributions, Maximum A Posteriori (MAP) inference
- Bayesian vs. frequentist interpretation of probability
- Practical session: Simple Bayesian inference exercises with Python (PyMC, Scipy.stats)
- Course Summary and exam review session

Software Tools:

- Python (Jupyter Notebook or similar IDE environment)
- NumPy, Pandas, Matplotlib, SciPy, Scikit-learn libraries.

Recommended References

- "Mathematics for Machine Learning" by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.
 - "Linear Algebra and Learning from Data" by Gilbert Strang.
 - "An Introduction to Statistical Learning" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
 - "Probability and Statistics for Computer Scientists" by Michael Baron.
-